

A Study on Apple Stock Price Behaviour *

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Abstract

The Stock Market is highly volatile and there are significant challenges in fitting a good model on a stock market data. This project is an attempt to study the closing price of Apple Stock data for the last 10 years, Jan. 2010 to Dec. 2019. We use the tools of Time Series analysis to analyze the data. Firstly, we suitably decompose the time series data to obtain its deterministic components and consequently remove the non-stationarity present in the data. Henceforth, we move on to find an appropriate model to model the stationary part, particularly using ARIMA methodologies. Finally, we forecast future values based on past observations to check the accuracy of the model.

Introduction

The stock market is very unpredictable, any geopolitical change can impact the share trend of stocks in the share market, recently we have seen how COVID-19 has impacted the stock prices, which is why on financial data doing a reliable trend analysis is very difficult. This is why it has always drawn the interest and attention of various professionals in the field of science or commerce. The randomness and forever fluctuating market makes investing a risky business where blindly following the trend or even the news, often intended to fool and misguide, can be devastating as was the case of Pfizer vaccine. In order to get ahead of the market, statisticians have always tried to analyse and forecast the stock price that in essence, reflects all known and unknown information in the public domain. In other words, **"Market discounts everything"**.

This project work is an attempt to perform **Time Series Analysis of the Apple Stock Price**. The motive of this project is to analyse the Apple stock closing price movement and possibly fit a suitable model to make forecasts. We have chosen closing price because it is the price at which the market settles down for a given day and can act as a reference price for the next day. The main difficulty lies in fitting a model to such a high frequency data.

The data used here is of the daily Apple stock market for 10 years, Jan 2010 to Dec 2019, and is extracted from the open source of Yahoo finance.

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Exploratory Data Analysis

This project work starts with the loading of the **univariate** Apple stock closing price ($\mathbf{X}_t$) from Yahoo finance official website from Jan 1, 2010 till Jan 1, 2020. We first begin by checking for presence of any deterministic part in the data.

For performing the test of randomness in the data we use the ‘**Turning point test**’.

Turning Point Test :

The test is based on the following hypotheses,

$\mathbf{H}_0$: Series is random

$\mathbf{H}_1$: Series is not random

In this case, we obtained a p-value of around $2.2e^{-16} < 0.05$.

Hence, we can conclude that our null hypothesis is rejected at 5% level of significance. i.e. the the time series is not random.

The plots of the original data ($\mathbf{X}_t$) and the log transformed data ($\log \mathbf{X}_t$) are presented below in **Fig.1** .

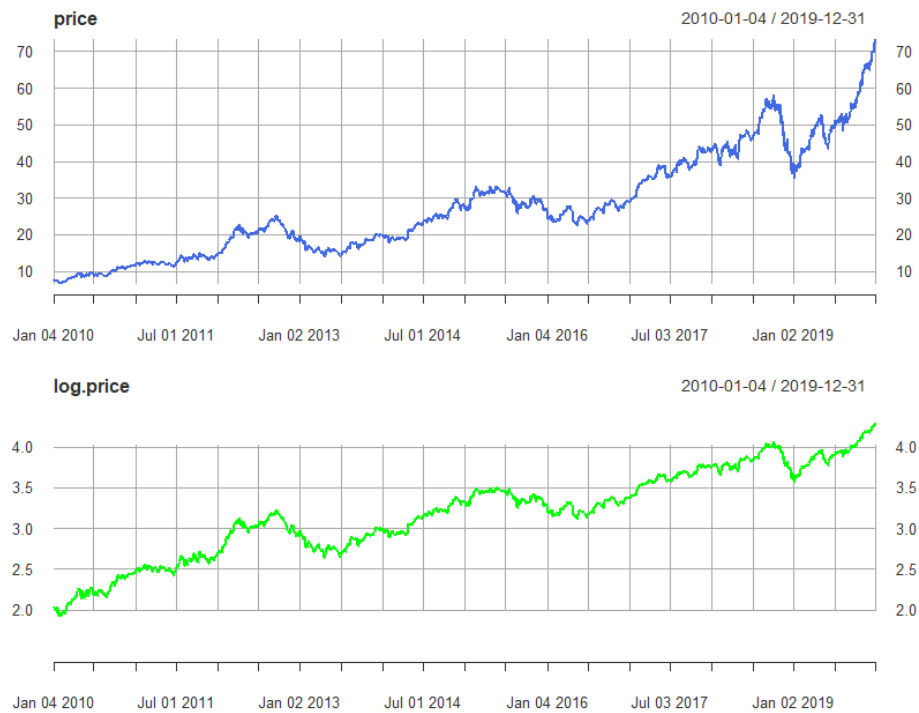


Figure 1: Closing price and log of closing price of Apple stock

By looking at both the plots, it is clear that both the plots have an increasing trend and are not white noise. The *log transformed data is considered for further analysis* because taking logarithm smooths the trend curve and in turn, helps to forecast easily. Thus our concerned model is give by:

$$\log \mathbf{X}_t = \mathbf{Y}_t = \mathbf{m}_t + \mathbf{s}_t + \epsilon_t$$

Data for the year 2019 is considered as test data and the rest is considered as training data. An attempt is made to forecast the stock price for the last 10 values of the year 2019. Keeping in mind that stock market is functional only during 252 days of the year, the train data is decomposed into its deterministic and stochastic components as shown in **Fig.2**.

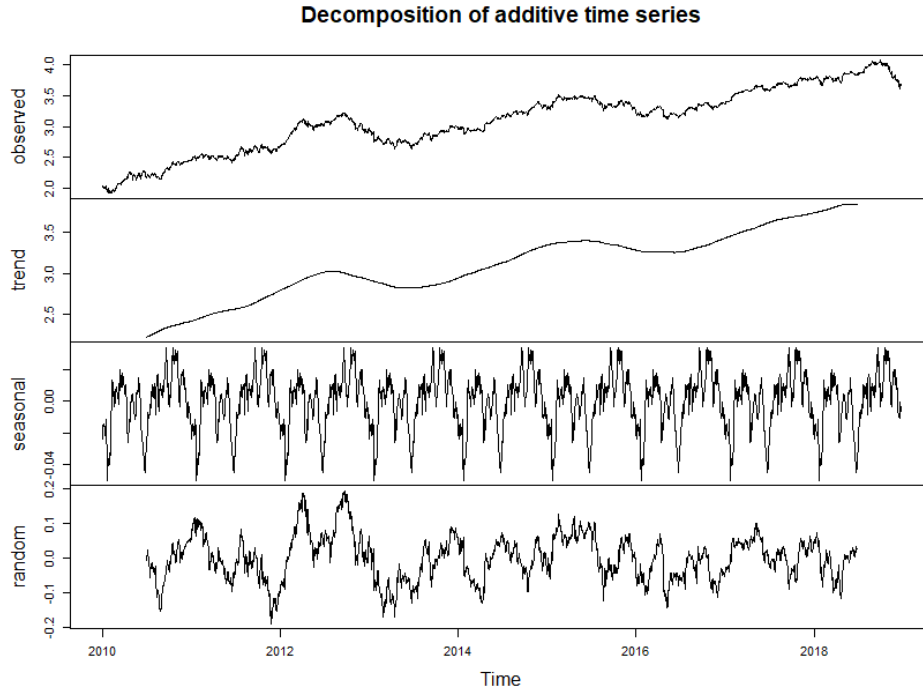


Figure 2: Decomposition of additive time series

Meticulous checking is done regarding the presence of each component separately. The presence of the seasonality in this high frequency data is highly unlikely though it shown in the decomposition plot. We need to investigate it.

Testing and Elimination of Deterministic components

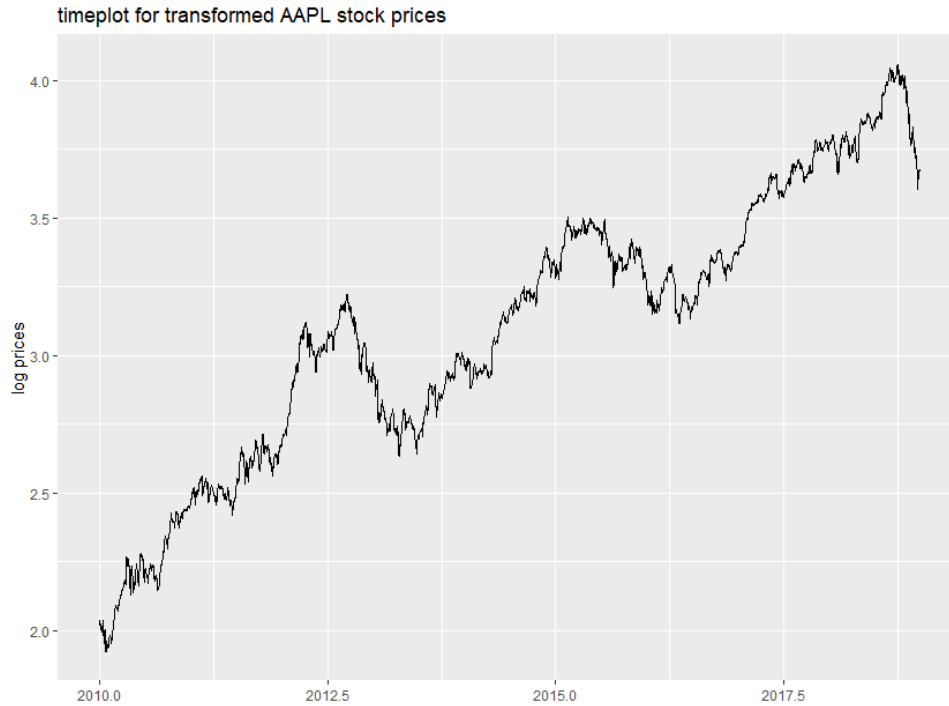


Figure 3: Time plot of the training data

From **Fig.3** it is evident that it has increasing trend and is **non-stationary**. First order differencing on the log transformed data is performed for elimination of trend and plotted in **Fig.4**. We can see that $\mathbf{Z}_t = \nabla \mathbf{Y}_t$ is more or less stationary .

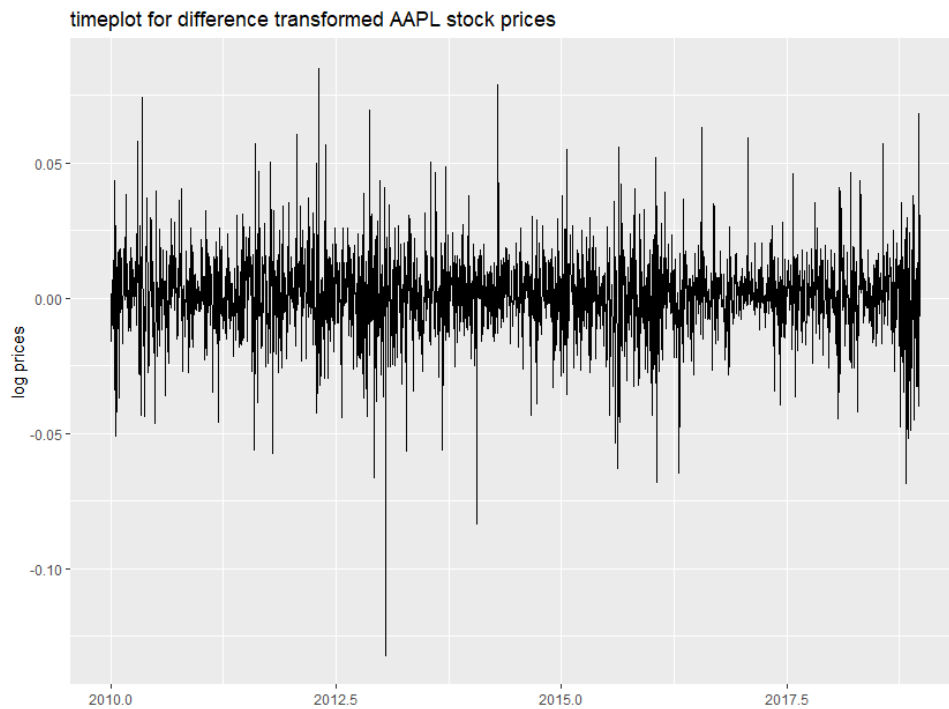


Figure 4: Plot of differenced log training data

Augmented Dickey-Fuller test:

The Augmented Dickey-Fuller test allows for higher-order autoregressive processes by including $\nabla \mathbf{Z}_{t-p}$ in the model. Our test is if $\gamma = 0$ in

$$\nabla \mathbf{Z}_t = \alpha + \beta t + \gamma \mathbf{Z}_{t-1} + \delta_1 \nabla \mathbf{Z}_{t-1} + \delta_2 \nabla \mathbf{Z}_{t-2} + \dots + \delta_p \nabla \mathbf{Z}_{t-p} + \epsilon_t$$

The test is based on the following hypotheses,

$\mathbf{H}_0$: Series is not stationary

$\mathbf{H}_1$: Series is stationary

The test results in the rejection of the null hypothesis since the p-value for the test come out to be very less, ensuring statistically that **first order differenced log transformed variable($\mathbf{Z}_t$) is stationary.**

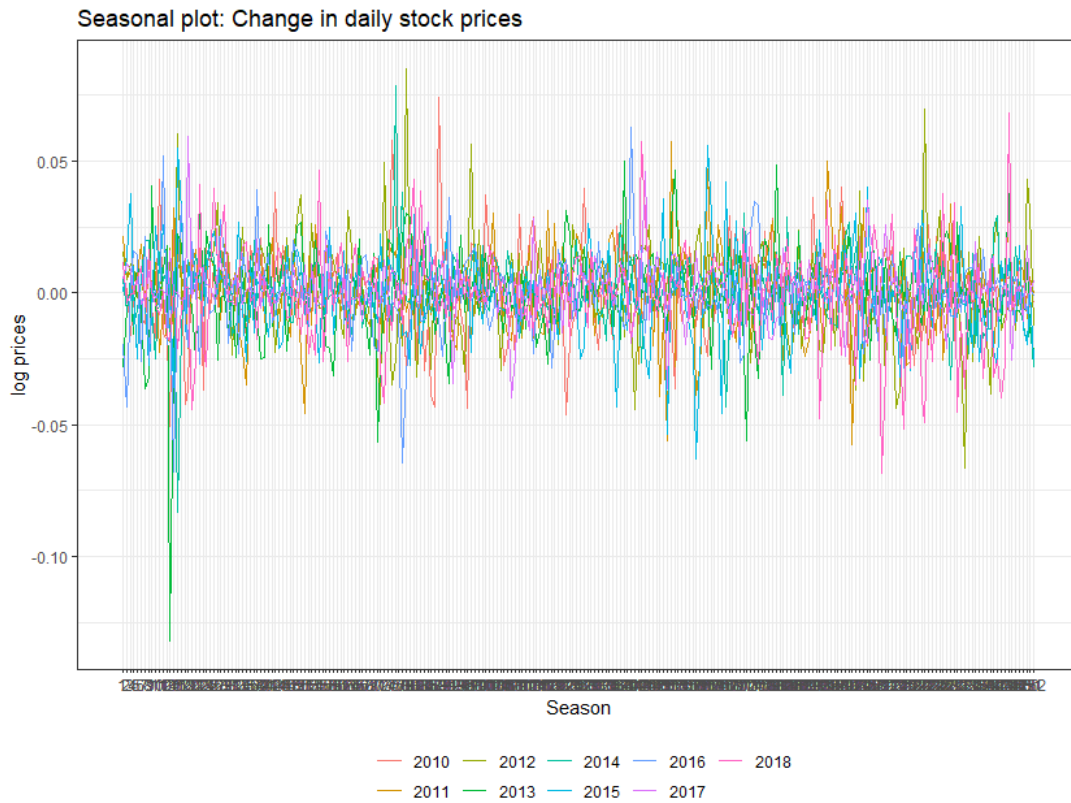


Figure 5:

We will now check our suspicion of no seasonality. In **Fig 5**, no definite pattern of seasonality can be seen in $\mathbf{Z}_t$.

To further substantiate our assumption, we perform **turning point test** on $\mathbf{Z}_t$. The P-value is $0.5385 > 0.05(\alpha)$. We may accept the null hypothesis H_0 and conclude that $\mathbf{Z}_t$ is stationary.

Model Selection

ACF and PACF Interpretation

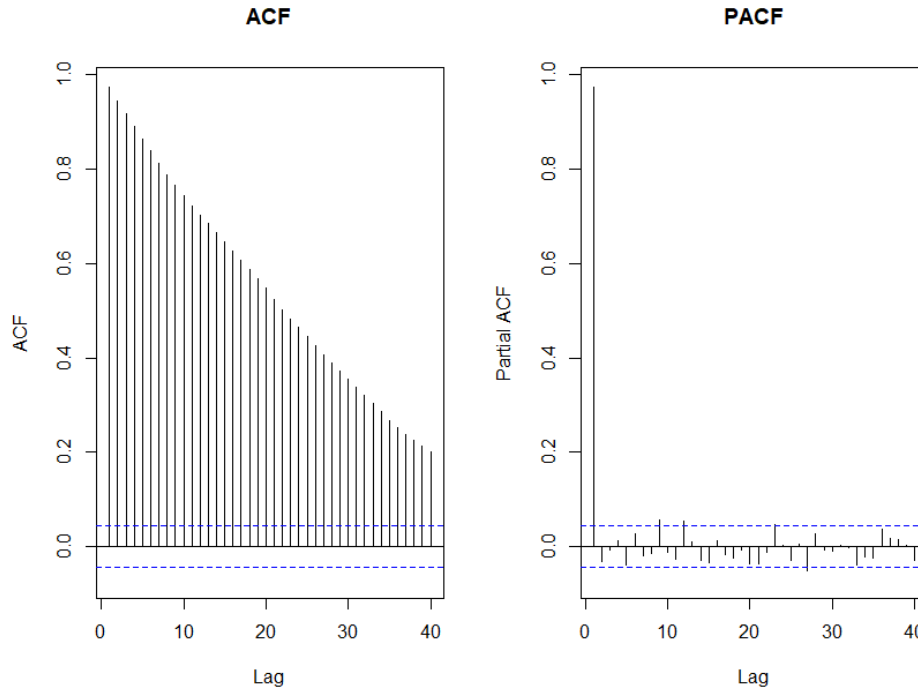


Figure 6: ACF and PACF plot

ACF and PACF plot of $\mathbf{Z}_t$ shows the correlation between the $\mathbf{Z}_t$'s at various lags

ACF - Autocorrelation refers to how correlated a time series is with its past values. For MA models we will use ACF plot to identify the (q) order.

Suppose we have time series data $\mathbf{A}_t$. If $\mathbf{A}_t$ is correlated with $\mathbf{A}_{t-1}$, and $\mathbf{A}_{t-1}$ is equally correlated with $\mathbf{A}_{t-2}$, then we should also expect to find correlation between $\mathbf{A}_t$ and $\mathbf{A}_{t-2}$. In fact, the amount of correlation we should expect at lag 2 is precisely the square of the lag-1 correlation. Thus, the correlation at lag 1 "propagates" to lag 2 and presumably to higher-order lags.

Similarly, here also the autocorrelations are significant for a large number of lags, but perhaps the autocorrelations at lags 3 and above are merely due to the propagation of the autocorrelation at lag 1 or 2. Thus, we need to investigate further and so we look at the PACF plot.

PACF - A partial autocorrelation is the amount of correlation between a variable and a lag of itself that is not explained by correlations at all lower-order-lags. By mere inspection of the PACF we can determine how many **AR** terms, we need to use to explain the autocorrelation pattern in a time series: if the partial autocorrelation is significant at lag k and not significant at any higher order lags i.e., if the PACF "cuts off" at lag k , then this suggests that we should try fitting an autoregressive model of order k .

If we look the PACF plot, we can note that it has a significant spike only at first lag and slightly significant another spike at lag 8 and 12, meaning that all the higher-order autocorrelations are effectively explained mostly by the first lag autocorrelation and at max, by the lag 8 or 12 autocorrelation.

Fitting the Appropriate Model

The ACF and PACF plots suggest that the time series is neither a MA nor an AR process and consequently an ARMA model is to be fitted to $\mathbf{Z}_t$.

The **ARMA(p,q)** model fitted on $\mathbf{Z}_t$, for which the (p,q) pair gives the minimum AIC value is chosen. Thus we check the AIC value of different ARMA models with the maximum value of p and q set at 8. We are setting the maximum value of p and q to be 8 because we don't want to form a complex model and also to prevent overfitting. We found out that the best fitted model is **ARMA(7,7)**.

The best fitted model is **ARMA (7,7)**:

$$\begin{aligned}\mathbf{Z}_t = & -0.3168\mathbf{Z}_{t-1} - 1.1509\mathbf{Z}_{t-2} + 0.0821\mathbf{Z}_{t-3} - 1.1999\mathbf{Z}_{t-4} - 0.0758\mathbf{Z}_{t-5} - 0.5630\mathbf{Z}_{t-6} \\ & + 0.4512\mathbf{Z}_{t-7} + \epsilon_t + 0.3469\epsilon_{t-2} + 1.1603\epsilon_{t-2} - 0.0619\epsilon_{t-3} + 1.2199\epsilon_{t-4} \\ & + 0.0956\epsilon_{t-5} + 0.6014\epsilon_{t-6} - 0.4320\epsilon_{t-7}\end{aligned}$$

where $\text{Var}(\epsilon) = 0.0002581$.

Thus, the best model fitted on $\mathbf{Y}_t$ is **ARIMA (7,1,7)**.

Residual Analysis

Residuals refer to the part of information in training or validation set that was not explained by the model fitted. Residual analysis is performed to validate if errors follow White Noise process. The auto-correlation function having values within the confidence interval of the corresponding estimates, will indicate that the residuals are uncorrelated.

We can also check that residual at time t is independent of past input variables X_i 's $i = 1 \dots (t - 1)$. Evidence of correlation indicates that the model does not describe how part of the output (Y_t) relates to the corresponding input. For example, any peak outside the confidence interval for lag k means that the output Y_t that originates from the input X_{t-k} is not properly described by the model.

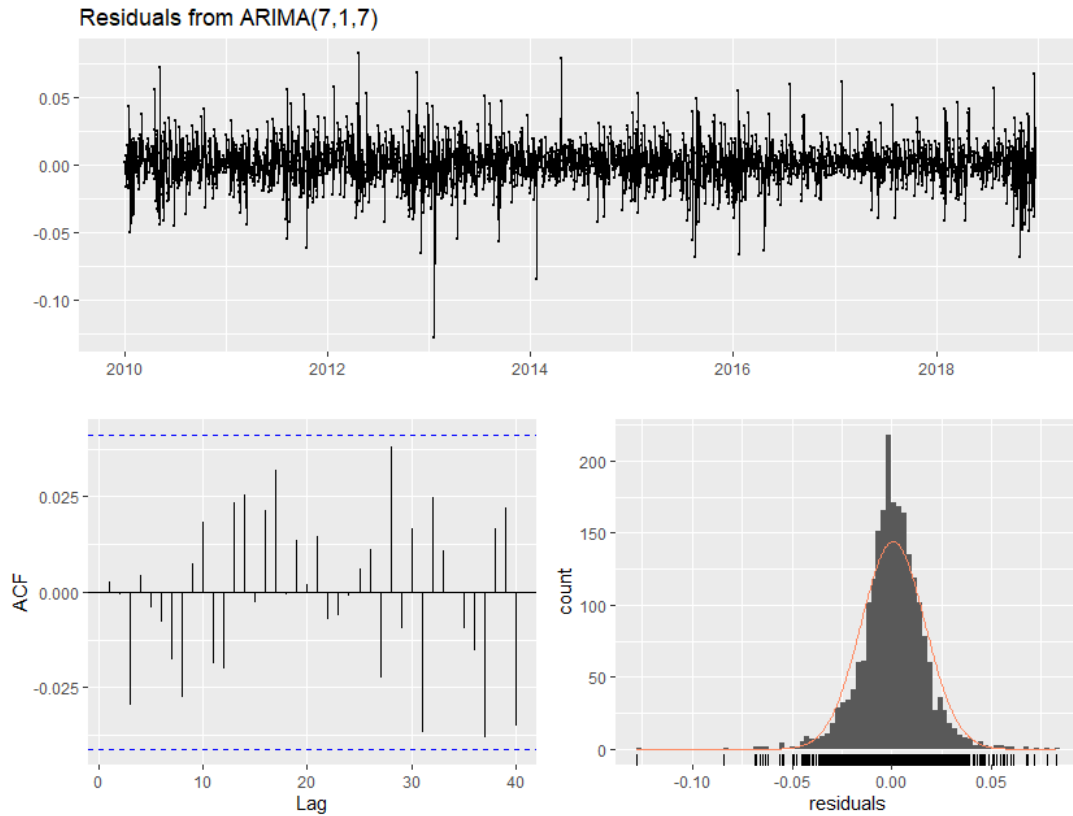


Figure 7: Residual analysis for fitted ARIMA (7,1,7)

From the histogram plot of the residuals in **Fig. 7**, we observe that the residuals are concentrated at 0 which suggests that the **mean** of the residuals is close to zero i.e, we have,

$$E(\epsilon_t) = 0 \quad \forall t$$

The time plot of the residuals shows that the variation of the residuals stays much the same across the historical data, apart from a few outliers, and therefore the residual **variance** can be treated as constant i.e, we have,

$$Var(\epsilon_t) = constant \quad \forall t.$$

Now, we wish to check if the errors are uncorrelated. From the ACF plot in **Fig. 7**, it is

clear that the errors are **uncorrelated**. Thus, we have,

$$Cov(\epsilon_t, \epsilon_{t+h}) = E(\epsilon_t, \epsilon_{t+h}) = 0 \quad \forall h > 0.$$

Thus our model assumption of the residuals being white noise is satisfied.

Furthermore, we perform another test, namely Ljung Box test to check if the errors are additionally independently distributed.

Ljung Box Test:

The Ljung–Box test is based on the following hypotheses,

H₀: The data are independently distributed.

H₁: The data are not independently distributed; they exhibit serial correlation.

The p-value of the test came out to be 0.4082 (≥ 0.05). Thus we fail to reject **H₀** at 5% level of significance. Hence, we can conclude that the residuals are also independent.

Consequently, forecasts from this model will probably be quite good, and prediction intervals that are computed assuming a normal distribution of the errors (an assumption made from the histogram) may also be accurate enough.

Forecasting

Now that an appropriate model is fitted, our next objective is to check how efficiently this model can predict future values, given the knowledge of past observations. In our study, we used the first 80% of the data to fit the model. For forecasting, we used the remaining 20% (minus 10) of the data to present the forecast values for the latest 10 points and compare them with actual observations.

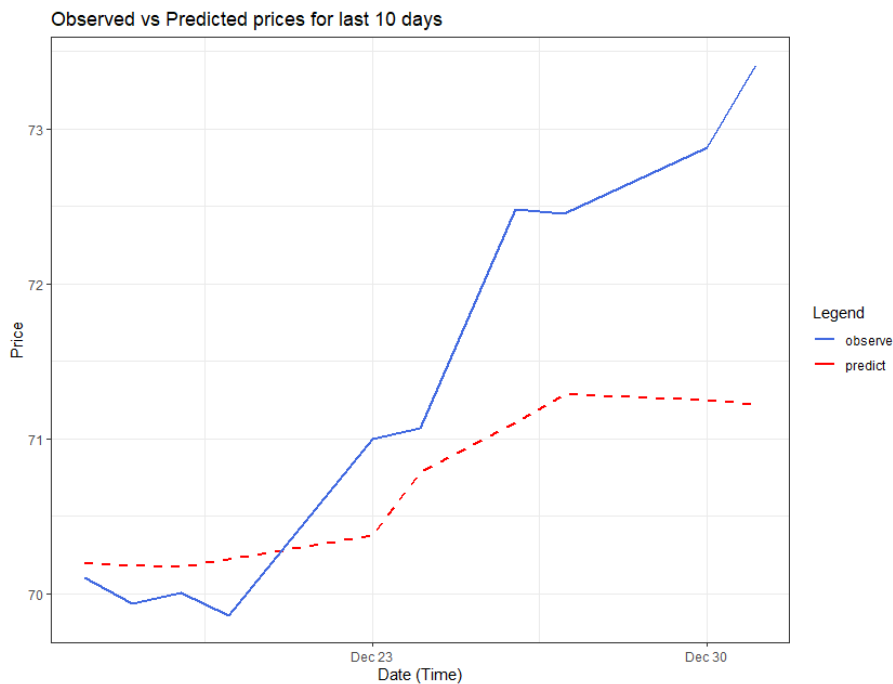


Figure 8: Forecast vs Actual observations for the latest 10 time points

We obtain a **Mean Square Error of 1.1443** for this forecasting. We are satisfied with this low value and thus complete our analysis.

Conclusion

The purpose of this project was to observe the predictability in the stock prices of Apple, i.e. if the history tells us anything about its future movement. And if positive, then find the most suitable model and approximate values for time in future. We have established that the process is not ideally random. Of the deterministic components, there exists only trend and no seasonality as one would expect for the stock prices. Cyclical, however, is not observable in the chosen length of data.

We conclude that **ARIMA(7,1,7)** is the most suitable fit for transformed (log) prices. We then forecast the price for the last 10 days based on fitting the model on remaining days of year 2019. The predicted values are close to the observed, yielding very less mean square forecast error, thus validating our claim for the model.

Reference:

- Class Notes of MTH 517A
- Wikipedia
- <https://people.duke.edu/~rnau/411arim3.htm>

Appendix

R-Code

```
rm(list=ls())
library(quantmod)
library(tseries)
library(timeSeries)
library(forecast)
library(xts)
library(fpp)
library(tsbox)
library(ggplot2)
library(ggfortify)
library(astsa)
library(spgs)

## getting AAPL data and subsequently it's closing price
getSymbols('AAPL', from='2010-01-01', to='2020-01-01')
price = AAPL[, 4]
log.price = log(price)
turningpoint.test(ts(log.price)) ##test for randomness

## plotting log and original dataset
par(mfrow= c(2,1))
plot(price, col= "royalblue"); plot(log.price, col= "green", ylim = c(1.5,4.5))

## Train-test split
log.price.train = log.price [c("/2018")]
log.price.test = log.price [c("2019/")]

## decomposing the data
sum( lubridate::year( index(AAPL) ) == 2010 )
## No. of observations for each year = 252
y = ts(log.price.train, frequency=252, start=c(2010, 1))
y.test = ts(log.price.test, frequency =252, start= c(2019,1))

n.train = length(y)
n.test = length(y.test)
y_components = decompose(y, )
plot(y_components)

autoplot(y) +
  ggtitle("timeplot for transformed AAPL stock prices") +
  ylab("log prices")
## Checking Stationary of log data (y)
adf.test(y, alternative = "stationary")

## want to investigate transformations (diferencing)
```

```

ndiffs(y) ## Take the first difference
dy = diff(y)
autoplot(dy) +
  ggtitle("timeplot for difference transformed AAPL stock prices") +
  ylab("log prices")
## series appears trend stationary

## Now investigate seasonality
ggseasonplot(dy) +
  ggtitle("Seasonal plot: Change in daily stock prices") +
  ylab("log prices") + theme_bw() +
  theme(legend.title = element_blank(), legend.position = "bottom")
## NO seasonality observed
## as each year behaves in different fashion
turningpoint.test(ts(dy))

##checking stationarity of data after removing deterministic components
y.random = y - y_components$trend
# summary(y.random)
y.random = na.omit(y.random)
autoplot(ts(y.random, frequency= 252, start = c(2010,1))) +
  ggtitle("Timeplot for stochastic component of time series") +
  ylab("random comp. in additive model") + theme_bw()
## Seems like only random behaviour

adf.test(y.random, alternative = "stationary")
## Reject Ho. => random comp. of data is stationary at 95% loc
## (H1 accepted)

## ACF and PACF of the error
par(mfrow=c(1,2))
Acf(y.random, lag.max = 40, main='ACF')
Pacf(y.random, lag.max = 40, main = 'PACF')
# AR value to 8 is sufficient

## Fitting ARIMA Model
best.order = c(0, 0, 0)
best.aic = Inf
for (q in 1:8) for (p in 1:8) {
  fit.model= arima(y, order = c(p, 1, q), optim.control = list(maxit = 1000) )
  fit.aic = fit.model$aic
  if (fit.aic < best.aic) {
    best.order = c(p, 1, q)
    best.arima = fit.model
    best.aic = fit.aic
  }
}

best.order; best.aic
## [1] 7,1,7

```

```

best.arima

##Final fit before Ljung test
model.fit = best.arima
# tsdisplay(residuals(model.fit), lag.max = 40,
main ="ARIMA (7,1,5) Model Residuals")
checkresiduals(model.fit, lag.max = 40)
Box.test(model.fit$residuals , lag=27 ,type = "Ljung-Box")

## Forecast
par(mfrow = c(1,1))

f = sarima.for(y.test[1:(n.test-10)], n.ahead =10, p=7, d=1, q=7); dev.off()

clrs = c( "royalblue", "red")
autoplot(ts(f$pred, frequency = 252, start =c(2019, 243) )) +
  autolayer(ts(f$pred, frequency = 252, start =c(2019, 243) ),
    series = "Predicted", linetype = "dashed", size=1)+
  autolayer(ts(y.test[150:252], frequency = 252, start = c(2019,150)),
    series="Observed" ) +
  ggtitle("Observed vs Predicted log prices") + theme_bw() +
  guides(colour=guide_legend() ) + scale_color_manual(values=clrs) +
  xlab("Year (Time)") + ylab("log prices") + xlim(2019.58,2020)+
  theme(legend.title = element_blank(),legend.position = "bottom")

dat = data.frame(date=index(tail(AAPL,10)),
  predicted = exp(f$pred), observed= exp(y.test[243:252]) )
colss=c("observe"="royalblue", "predict"="red")

ggplot(dat, aes(x= date)) +
  geom_line(aes(y= predicted, color = "predict"), linetype="dashed", size=1) +
  geom_line( aes(y=observed, col = "observe"), size=1) +
  labs(title="Observed vs Predicted prices for last 10 days" ,
    x="Date (Time)", y="Price", color="Legend") +
  theme_bw() + scale_color_manual(values = colss)

##MSE for forecasting
p = exp(f$pred)
obs = exp(y.test[243:252])
MSE_foreast = mean((p - obs)^2)
MSE_foreast

```